Largest Semiconductor Products Manufacturer

→ Nvidia

Largest Electronics Products Company

→ Apple (By Market Cap)

Jensen Huang
→ CEO (Nvidia)

Lisa Su
→ CEO (AMD)

Morris Chang
→ Founder (TSMC)

Cristiano Amon
→ CEO (Qualcomm)

Sanjay Merhotra
→ CEO (Micron)

JC Bose, credited in Shockley's speech for making the first transistor

IDM v/s Fab v/s Design

Correlation b/w electronics & semicon industry in india

Elec	200B	500B	3T
Semicon	40B	135B	810B
	2023	2030	2047

5.8% of GDP

10.2% of GDP

Products & Technology are essential for properly growing economy & strategic autonomy.

90% consumption from Global brands from electronics PoV

→ Technology soverignity (can be shut-off anytime)

→ Economic Self-reliance

(Allows us to reinvest our country's consumption in our RnD)

→ Components ecosystem

(If we do products, our manufacturers will also have a larger viable market.)

Electronics Products Growth Domains

1) AI / ML ***

2) Auto / EV

3) Consumer Electronics

4) Telecom & Wireless

5) Medical Bio-Tech

6) Energy (Clean Tech)

7) Space Defense

AI & Semiconductor

inter-connection

1980's	Now
Required supercomputers to simulate as small as 500-node NN's.	Pocket-sized devices managing LLM's w/ trillions of parameters

→ Emerging Trends "7"

- Security
- Custom Chips (Google / Microsoft / NVIDIA / Amazon)
- Architecture Explosion
- Advanced 3D Packaging & Chiplets
  - ↳ We need vertical integration

Photomask & Reticles

↳ Learn work & dimensions

AI causing them to reach limit

- Wireless functionality
- Photonics based interconnect b/w chips
- Heterogeneous & vertical integration

Manufacturing

→ Electronics, EMS
→ Semicon fab, AMTP, etc.

Products

→ Electronics
→ Chips & IC's

Quality Talent & Research

Quality Education → Outcome-Based Research → Deployable Technologies → Commercial Products

Industry → Academia

Semiconductor Industry

Semiconductor Chips

Manufacturing

Wafer

Digital Design

Analog Design

Devices

fab packaging

NVIDIA, Qualcomm, etc.

IMEC, IBM, TSMC, Micron, etc.

→ RTL Design
→ Verification
→ Physical & Digital Design

Build fundamental understanding using complicated projects rather than remaining pedagogy focused

~~32-bit adder~~ Make a whole chip

X-86(Intel), ARM → Proprietary Architecture

RISC-V → Open-source, customisable architecture
